

- sockets package facilitates TCP/IP and UDP communication, which are used to enable networking in IoT devices.
- Reference: <https://pypi.org/project/sockets/>
- *paho-mqtt*: Install using 'pip install paho-mqtt'.
 - The MQTT or Message Queue Telemetry Transport protocol is a publish/subscribe messaging transport that is intended for machine-to-machine (M2M)/Internet of Things connectivity, enabling high speed data exchange with low payload.
 - When a minimal code footprint is necessary or bandwidth on the network is at a premium, it is beneficial for communications with faraway locations.
 - Applications can connect to a MQTT broker to publish messages, subscribe to topics, and receive published messages thanks to the client class provided by the Paho library code.
 - Additionally, it offers certain assistance functions to make it simple to publish one-time messages to MQTT servers.
 - Reference: <https://pypi.org/project/paho-mqtt/>
- *smtplib*
 - This protocol handles the sending and routing of emails between mail servers.
 - It is part of Python's standard library.
 - Reference: <https://docs.python.org/3/library/smtplib.html>
- *asyncio*: Install with 'pip install asyncio'.
 - The asyncio module provides architecture for building single-threaded concurrent programming with co-routines, multiplexing I/O access through sockets and other resources, running network clients and servers, and other primitives.
 - Reference: <https://pypi.org/project/asyncio/>
- **IoT database**
 - *MySQLdb*: Install with 'pip install MySQL-python'.
 - MySQLdb is a popular go-to relational database that aids in the building of remote storage for IoT systems.
 - Reference: <https://pypi.org/project/MySQL-python/>
- *SQLite*
 - SQLite is a C library that provides a lightweight disk-based database that does not require a separate server process and may be accessed using a non-standard variation of SQL.
 - SQLite can be used by IoT to store internal data.
 - The sqlite3 module does not require installation. Since Python 2.5, it has been included in the standard library.
 - Reference: <https://docs.python.org/3/library/sqlite3.html>
- **IoT graphical user interface**
 - *Tkinter*
 - Python provides numerous GUI (graphical user interface) development choices. Tkinter is the most commonly used of all the GUI techniques.
 - It is a standard Python interface to the Python-supplied Tk GUI toolkit.
 - Python with Tkinter is the quickest and most straightforward approach to construct GUI apps.
 - Reference: <https://docs.python.org/3/library/tkinter.html>
- **IoT backend**
 - *Flask*: Install via 'pip install Flask'.
 - Flask is a simple micro framework for building web applications, to build http requests.
 - IoT applications require a web-based interface in order to provide monitoring and controlling devices.
 - Reference: <https://pypi.org/project/Flask/>
 - *websockets*: Install using 'pip install websockets'.
 - websockets is a Python package for creating WebSocket servers and clients. It prioritises consistency, simplicity, resilience, and performance.
 - The WebSocket API is a cutting-edge technology that allows a two-way interactive communication session to be established between a user's browser and a server. This API sends messages to a server and receives event-driven responses without polling the server for a response.
 - Reference: <https://pypi.org/project/websockets/>
 - *requests*: Install with 'pip install requests'.
 - The requests module allows sending HTTP requests; the HTTP request generates a 'response object' containing response data (content, encoding, status, and so on).
 - Supports multipart file uploads, streaming downloads, and connection pooling.
 - Enables automatic decoding and decompression of content.
 - Reference: <https://pypi.org/project/requests/>
 - *aiohttp*: Install with 'pip install aiohttp'.
 - aiohttp provides an asynchronous HTTP client or server framework, including websockets support.
 - Reference: <https://pypi.org/project/aiohttp/>
 - *pushsafer*: Install with 'pip install python-pushsafer'.
 - pushsafer can send and receive push notifications in real-time to iOS, Android, and Windows devices (mobile and desktop), as well as browsers such as Chrome, Firefox, Opera, and others.
 - Reference: <https://pypi.org/project/python-pushsafer/>
- **IoT data analysis and visualisation**
 - *Numpy*: Install via 'pip install numpy'.